Production RAG

User Query
(Input Question)

Retriever
(Related Docs)

Vector Store
(Embedded Documents Indexed for Search)

Context + Query
(Combined Prompt)

****RAG grounds LLM responses based on the actual documents which reduces hallucinations

LLM
(Generate Answer)

Response
(Grounded Docs)

BASIC RAG CHAIN

{context, question}
Parallel inputs

Prompt
(template)

LLM
(model)

Parser
(output)

Context ----> Retriever

Question ----> Runnable Passthrough

(Runnable passthrough is a term comes with langchain. It receives the input and carries it through unchanged)

Prompt Template

Answer based on the {context}
Question: {question}

Handling "I don't know scenarios"

i. Without proper instructions → (will make up answer for question)
ii. With proper instructions → (will answer as I don't know)

So, pattern of the prompt something like "answer based on the context only if the context does not contain the answer, say I don't know"

RAG with sources.

Retriever output will have the informative record of the source from which page or doc got this.

Can do this by formatting the docs with sources. Adding source tag for each chunk. → formatted sources.

Why it matters?

Users can verify answers and cite on them. Builds the trust.

Document Loaders

Raw Files -> .txt , .docx , .html , .pdf , .csv

Document loader of the langchain takes it as input, return a wrapper object dictionary

List[Document]

page_content → actual text content

metadata → source, page, author

Core Document Loaders

PyPdfLoader → loads .pdf file

TextLoader →.txt file

DirectoryLoader → load multiple files

WebBaseLoaders → for web pages

UnstructuredLoader → for complex docs like .md mixed

Loading Options

PyPdfLoader → fast, basic extraction {speed: good, metadata: basic} for simple pdfs

PyMuPdfLoader → fastest, good metadata {speed: best, metadata: rich} for high volume

UnstructuredPdfLoader → best for complex layouts {speed: slower, metadata: detailed}

Folder/directory → DirectoryLoader {path: directory, glob: **/*.pdf, loader_cls: PyPdfLoader}

glob is pattern filter files (/*.pdf means all the pdfs including subfolders of this directory)